

Name _____ Index No. _____ / _____

2920/201
SYSTEMS ANALYSIS AND DESIGN
November 2015
Time: 3 hours

Candidate's Signature _____

Date _____



THE KENYA NATIONAL EXAMINATIONS COUNCIL
DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY
MODULE II
SYSTEMS ANALYSIS AND DESIGN
3 hours

INSTRUCTIONS TO CANDIDATES

Write your name and index number in the spaces provided above.

Sign and write the date of examination in the spaces provided above.

*Answer any **FIVE** of the following **EIGHT** questions in the spaces provided on this question paper.*

All questions carry equal marks.

Candidates should answer the questions in English.

For Examiner's Use Only

Question	1	2	3	4	5	6	7	8	Total Score
Candidate's score									

This paper consists of 16 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

1. (a) Outline **four** roles of a systems analyst as a change agent.

(4 marks)

(b) Explain **two** characteristics of deterministic systems.

(4 marks)

(c) Distinguish between *waterfall* and *phased* approaches as used in system development.

(4 marks)

(d) A well-designed user documentation would yield better results when used in an organization. Explain **four** good qualities that could be in this documentation. (8 marks)

2. (a) Define each of the following terms as used in system analysis and design:
- (i) sub-system; (2 marks)
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-
-
- (ii) environment. (2 marks)
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-
-
- (b) Explain **two** economic challenges experienced during systems development. (4 marks)
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-
-
-
- (c) Differentiate between *State Transition Diagram* (STD) and *Entity Life History* (ELH) as used in system design. (4 marks)
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-
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-
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-

- (c) Differentiate between *HIPO charts* and *IPO charts* as used in information systems. (4 marks)

[illegible]

- (d) Figure 1 shows a closed loop control system used in information systems. Explain the functions of the parts labelled (i), (ii), (iii) and (iv). (8 marks)

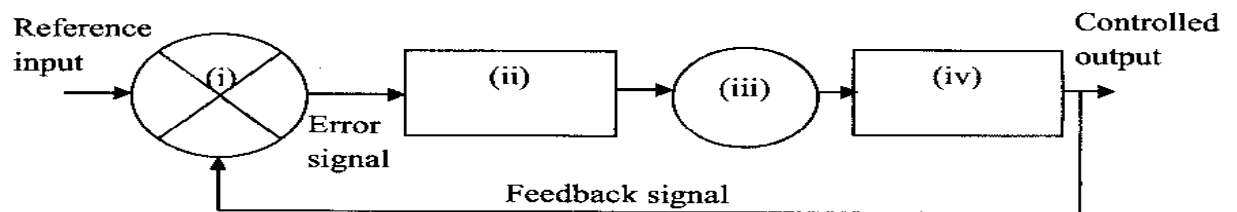


Figure 1

[illegible]

4. (a) Outline **four** limitation of a prototyping as used in system design. (4 marks)

- (b) Explain **two** disadvantages of *direct* changeover technique as used in systems implementation. (4 marks)

- (c) With the aid of sketches, explain **two** types of relationships used in ER diagrams. (4 marks)

- (d) The Strategic Plan Steering Committee of Hazel Ltd. intends to acquire a new information system for the organization. Explain **four** factors that should be considered when choosing the system. (8 marks)

5. (a) Outline **four** limitations of telephone interview as a data collection tool. (4 marks)

- (b) Explain **two** issues addressed by the operational feasibility study of an information system. (4 marks)

- (c) Distinguish between *black box testing* and *white box testing* as used in systems development. (4 marks)

(d) With the aid of a diagram, outline the stages of Soft System Methodology. (8 marks)

1

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6. (a) Explain **two** benefits that could be derived from using an adaptive information system. (4 marks)

- (b) Distinguish between *Jackson Structured Programming* (JSP) and *Jackson System Development* (JSD) as used in systems development. (4 marks)

- (c) Jewel Company Ltd. intends to train its employees in order to improve the overall performance of the organization. Explain **two** types of users that could be trained. (4 marks)

- 1

- [illegible]

- (b) Explain **two** disadvantages of payback analysis as used in systems analysis and design. (4 marks)

- (c) Explain **two** circumstances under which record inspection could be most appropriate in data collection. (4 marks)

- (d) Table 1 shows activities of a building construction project. Using the Critical Path Method, represent the project activities on a network diagram showing the EST and LST for each activity. (8 marks)

Task	Preceding task	Duration
A	-	5
B	A	4
C	A	3
D	C	7
E	D	6
F	B	9
G	F	5
H	G, E	3

Table 1

8. (a) Distinguish between *usability* and *readability* as used in systems design. (4 marks)

[illegible]

- (b) In the development of an information system, sometimes the project is derailed due to other external factors. Outline **four** factors that could lead to project derailment. (4 marks)

- (c) Systems review is a significant step for an information system staff to carry out. Explain **two** roles that could be played by this activity. (4 marks)

- (d) The following is information about a tour agent who deals in bus and hotel reservations. Use it to answer the question that follows.

A tour agency arranges holidays for clients. Reservations are made directly by clients. When a client makes an approach, the Client Service clerk selects appropriate bus details and hotel details from a list which are regularly updated. The details are entered onto a Reservation file. The client must confirm these reservations within three days by sending a deposit of 40% of the costs. On receipt of the deposit, the reservation office transfers the details from the Reservation file to the Confirmed Booking file. Two weeks before the bus is due, Accountant sends an invoice to the client for the remaining costs. Accounts notify Client Service when the full payment is received and Client Service then sends tickets and joining instructions to the client.

Draw a level 1 DFD to represent this information.

(8 marks)

This image shows a single sheet of white paper with horizontal blue or grey ruling lines, typical of notebook paper. The lines are evenly spaced and run across the width of the page. There is a small, dark speck near the top center of the page. The paper appears slightly aged or off-white.